

# Version Control and Collaboration -- Guided Notes

## ANSWER KEY

Unit tp-4.4 | CompTIA Network+ N10-009 Obj. FC0-U71 4.4 | N10-008 FC0-U71 4.4

For instructor use / distribute after students complete the notes.

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### Question 1

Version control (also called source control or revision control) is a system that tracks \_\_\_\_\_ to code over time. It allows developers to \_\_\_\_\_ to any previous state, compare changes, and work \_\_\_\_\_ without overwriting each other's work.

*Answer: Changes (edits); revert; collaboratively.*

### Question 2

Git is a \_\_\_\_\_ version control system, meaning every developer has a \_\_\_\_\_ copy of the full repository history. SVN is a \_\_\_\_\_ version control system, meaning there is one central server. With Git, developers can \_\_\_\_\_ without internet access and sync later.

*Answer: Distributed; complete (local); centralized; commit (work).*

### Question 3

A \_\_\_\_\_ is a snapshot of the repository at a point in time. When a developer saves changes, they create a commit with a \_\_\_\_\_ describing what changed. A \_\_\_\_\_ is a pointer to a specific commit and typically named for the development stream (e.g., main, develop, feature-login).

*Answer: Commit; message; branch.*

### Question 4

A \_\_\_\_\_ request (also called a merge request) is a formal proposal to merge code from one branch into another. It triggers a \_\_\_\_\_ review where team members inspect the code before it is merged. This process catches bugs and enforces \_\_\_\_\_ standards before defects reach the main branch.

*Answer: Pull; code; quality (coding).*

### Question 5

A merge conflict occurs when two developers have edited the \_\_\_\_\_ lines of the same file in different branches. Git cannot automatically decide which version is correct, so it \_\_\_\_\_ the conflict and asks the developer to \_\_\_\_\_ resolve it. Frequent small commits reduce the severity of merge \_\_\_\_\_.

*Answer: Same; flags (marks); manually; conflicts.*

### Question 6

GitHub, GitLab, and Bitbucket are \_\_\_\_\_ platforms that host Git repositories. They add features like \_\_\_\_\_ (tracks bugs and feature requests), \_\_\_\_\_ (automated testing pipelines), and team \_\_\_\_\_ (permissions, branch protection rules) on top of core Git.

*Answer: Cloud-based; issue tracking; CI/CD pipelines; management.*

### Question 7

A .gitignore file tells Git to \_\_\_\_\_ tracking specific files or directories. Common entries include \_\_\_\_\_ (compiled output), \_\_\_\_\_ (local environment configs with secrets), and \_\_\_\_\_ (IDE configuration directories). Committing secrets to a public repo is a major \_\_\_\_\_ risk.

*Answer: Stop (ignore); build/dist folders; .env files; .idea / .vscode; security.*

### Question 8 -- Real World Application

REAL WORLD: A student accidentally commits their database password directly into their public GitHub repository in a config file. They realize it within 10 minutes and delete the file from the repo. Explain why deleting the file is NOT sufficient to protect the secret and what they must do to remediate the situation.

*Answer: Deleting the file is not sufficient because Git stores the entire history of every commit. Even after deletion, the password is still present in the commit history and anyone who cloned or browsed the repository before or after the deletion can access it via git log, git show, or by viewing earlier commits on GitHub. Additionally, automated bots scan GitHub continuously for newly pushed secrets and can capture them within seconds of a push -- the 10-minute window may have already been too late.*

*Remediation: (1) Immediately invalidate (rotate) the compromised database password -- change it in the database right now, regardless of whether exposure is confirmed. The old credential must be treated as compromised. (2) Remove the secret from all git history using git filter-branch or the BFG Repo Cleaner tool, then force-push the cleaned history. (3) Going forward, never commit secrets -- use environment variables, .env files (added to .gitignore), or a secrets manager.*